Questão

1 $f(t) = t - \pi$ no intervalo $[0, 2\pi)$

$f(t) = f(t + 2n\pi)$, n é inteiro

1- $f(t) = f(t + 2n\pi)$ indica que a função é periódica.
Se a função é periódica, então podemos calcular os coeficientes da série de Fourier. → e obedece as condições de Dirichlet

$f(t) = f(t + T) = f(t + 2\pi n)$
Para o menor valor positivo $(n = 1)$, $T = 2\pi$

2 - Cálculo da série de Fourier

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi k \cdot t/T}$$

$$X_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, e^{-i2\pi k \cdot t/T}\, dt = \frac{1}{T} \int_0^T x(t)\, e^{-2i\pi k t/T}\, dt$$

→ Coeficiente da Série de Fourier

$x(t) = f(t) = t - \pi$

$$X_k = \frac{1}{T} \int_0^{2\pi} (t - \pi) \cdot e^{-i2\pi k \cdot t/T}\, dt$$

$$X_k = \frac{1}{T} \left\{ \underbrace{\int_0^{2\pi} t\, e^{-2i\pi k \cdot t/T}\, dt}_{1} - \pi \cdot \underbrace{\int_0^{2\pi} e^{-2i\pi k \cdot t/T}\, dt}_{2} \right\}$$

① Integração por partes

$T = 2\pi \qquad \frac{-2i\pi k}{T} = -\frac{2i\pi k}{2\pi} = -ik$

$$\int_0^{2\pi} t \cdot e^{-ikt} \Rightarrow \int u\,v = u.v - \int v\,du$$

$u = t \quad du = dt \qquad dv = e^{-ikt} \quad v = \frac{e^{-ikt}}{-ik}$

$$\frac{t \cdot e^{-ikt}}{-i.k}\Bigg|_0^{2\pi} - \frac{1}{-ik} \cdot \int_0^{2\pi} e^{-ikt}$$

$$\frac{1}{-ik}\left\{ 2\pi \cdot e^{-ik2\pi} - \frac{1}{-ik}\left[e^{-ikt} \right]_0^{2\pi} \right\}$$

$$\frac{1}{-ik}\left\{ 2\pi \cdot e^{-ik2\pi} + \frac{1}{ik}\left[e^{-ik2\pi} - 1 \right] \right\}$$

$$\frac{1}{-ik}\left\{ e^{-ik2\pi}\left(2\pi + \frac{1}{ik} \right) - \frac{1}{ik} \right\}$$

$$ik \cdot -ik = k^2$$

$$\left\{ e^{-ik2\pi}\left(\frac{2\pi}{-ik} + \frac{1}{k^2} \right) - \frac{1}{k^2} \right]$$

$$e^{-ik2\pi} \cdot \left(\frac{-2\pi \cdot ik}{-ik \cdot ik} + \frac{1}{k^2} \right) - \frac{1}{k^2} = e^{-ik2\pi}\left(\frac{2\pi i k}{k^2} + \frac{1}{k^2} \right) - \frac{1}{k^2}$$

$$= \frac{e^{-ik2\pi}\cdot(2\pi i k+1)-1}{k^2} = \frac{(2\pi i k+1)-1}{k^2} = \frac{2\pi i \cancel{k}}{k^{\cancel{2}}}$$

$$= \frac{2\pi i}{k}$$

$$e^{-ik2\pi} = \underbrace{\cos(2k\pi)}_{1} - i\,\underbrace{\sin(2k\pi)}_{0}$$

(2) $$\pi \cdot \int_0^{2\pi} e^{-2i\pi k \cdot t/T}\,dt = \pi \cdot \int_0^{2\pi} e^{-ikt}\,dt$$

$$T = 2\pi \qquad \frac{-2i\pi k}{T} = \frac{-2i\pi k}{2\pi} = -ik$$

$$= \pi \cdot \frac{1}{-ik}\left[e^{-ikt}\right]_0^{2\pi} = \frac{\pi i}{k}\left[e^{-ik2\pi} - 1\right]$$

$$e^{-ik2\pi} = \underbrace{\cos(k2\pi)}_{1} - i\,\underbrace{\sin(k2\pi)}_{0}$$

$$= \frac{\pi i}{k} \cdot [1-1] = 0$$

Juntando ① e ②

$$X_K = \frac{1}{T}\left\{\int_0^{2\pi} t\, \underset{①}{e^{-2i\pi K \cdot t/T}}\, dt - \pi \cdot \int_0^{2\pi} \underset{②}{e^{-2i\pi K \cdot t/T}}\, dt\right\}$$

$$X_K = \frac{1}{2\pi} \cdot \left(\frac{2\pi i}{K} - 0\right) = \frac{i}{K}$$

$$\frac{1}{T}\int_0^{T} x(t)\, e^{-2i\pi K t/T}\, dt$$

$$x(t) = \sum_{K=-\infty}^{\infty} X(K) \cdot e^{i 2\pi K/T}$$

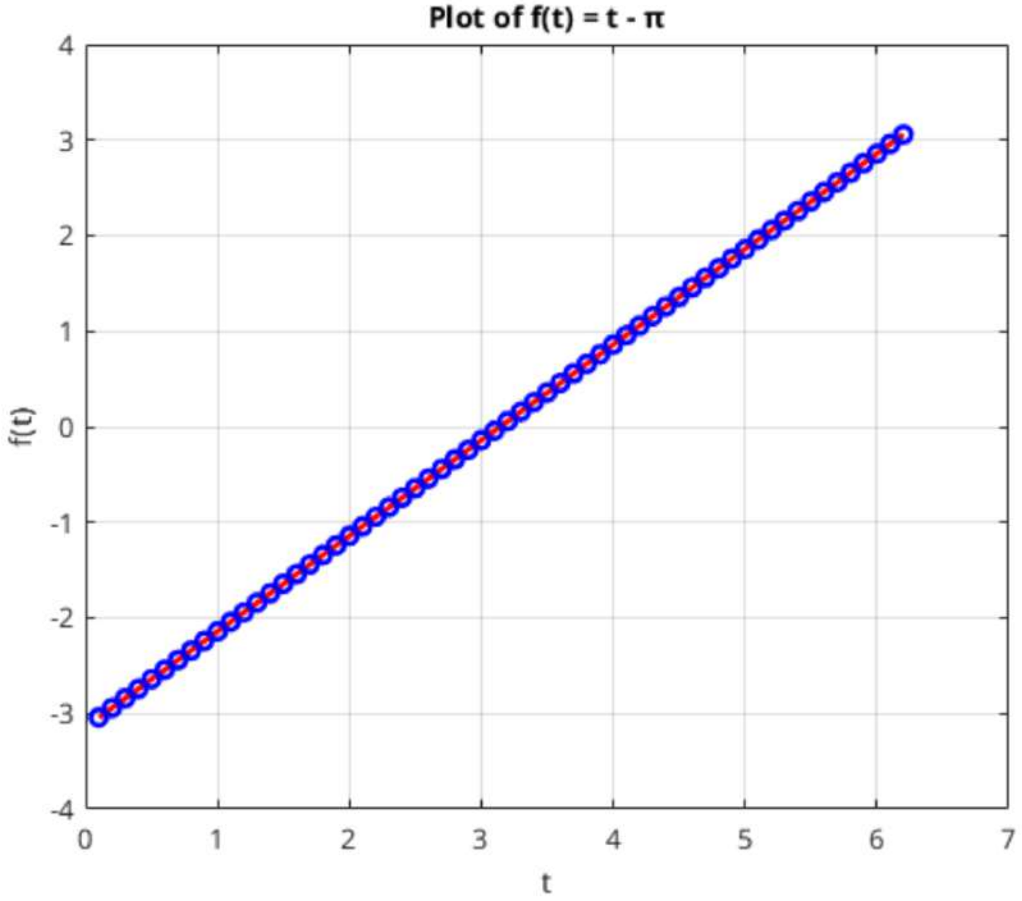
$$X_K = \frac{1}{T} \cdot \int_0^{T} x(t) \cdot e^{-i2\pi K t/T}\, dt$$

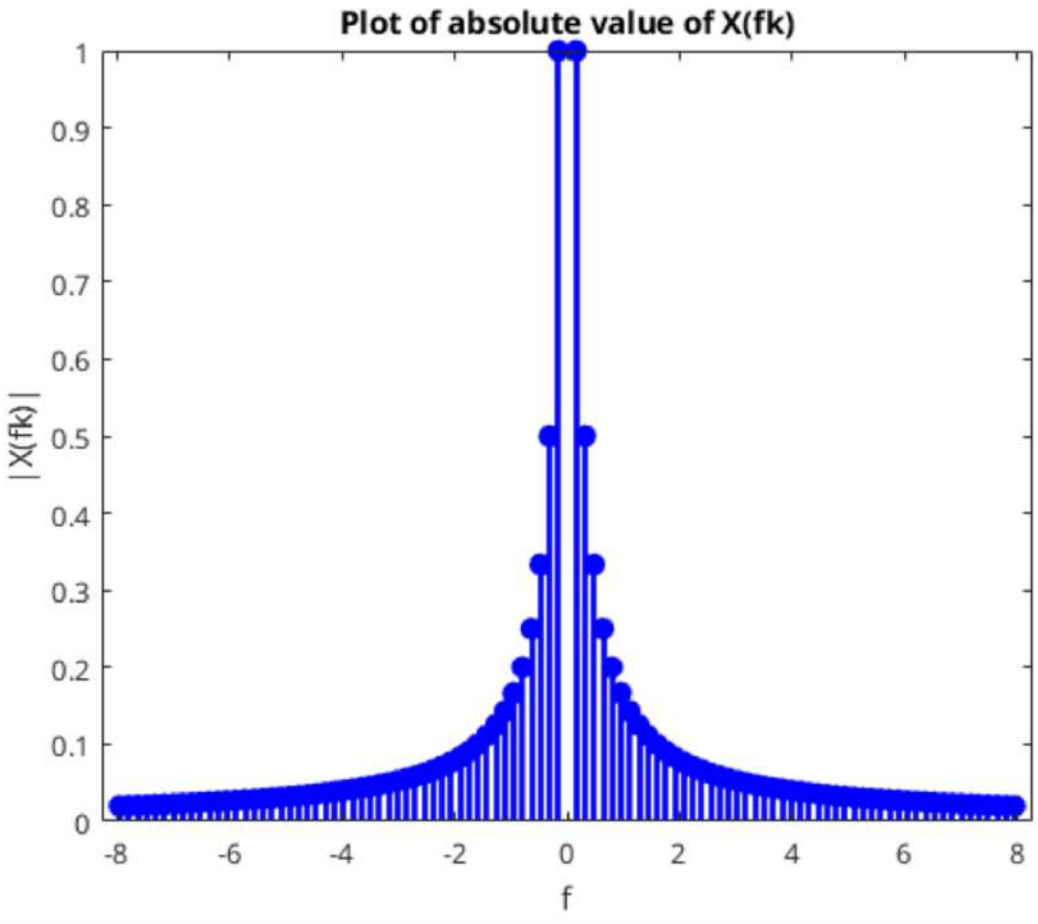
$$T = 2\pi$$

$$x(t) = \sum_{K=-\infty}^{\infty} X(K)\, e^{iKt} = \sum_{K=-\infty}^{\infty} \frac{i}{K} \cdot e^{iKt}, \quad K \neq 0$$

$$= i \cdot \sum_{K=-\infty}^{\infty} \frac{e^{iKt}}{K}$$

$$|X_K| = \sqrt{\left(\frac{1}{K}\right)^2} = \frac{1}{K}\sqrt{1} = \frac{1}{K}$$


Plot of f(t) = t - π
f(t)
t
4
3
2
1
0
-1
-2
-3
-4
0
1
2
3
4
5
6
7


Plot of absolute value of X(fk)
|X(fk)|
f
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-8
-6
-4
-2
0
2
4
6
8

Questão 2

1- $f(t) = \exp(t) = e^t$ intervalo 0 a 2π

$f(t) = f(t + 2n\pi)$, função periódica

$f(t) = f(t + T) = f(t + 2n\pi)$

$T = 2n\pi$, e para n inteiro e igual a 1
$T = 2\pi$

2- Calcule a série de Fourier, supondo que as condições de Dirichlet foram obedecidas.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \, e^{i2\pi k t/T}$$

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi k t/T} \, dt$$

Resolvendo a Integral

$$\int_0^{2\pi} e^t \cdot e^{-i2\pi k t/T}\, dt = \int_0^{2\pi} e^t \cdot e^{-i2\pi k t/2\pi}\, dt$$

$T = 2\pi$

$$\int_0^{2\pi} e^t \cdot e^{-ikt}\, dt = \int_0^{2\pi} e^{t(1-ik)}\, dt = \frac{1}{1-ik} \cdot \left[e^{t(1-ik)} \right]_0^{2\pi}$$

$$= \frac{1}{1-ik} \cdot \left[e^{2\pi(1-ik)} - 1 \right]$$

$$X_k = \frac{1}{2\pi(1-ik)} \left[e^{2\pi(1-ik)} - 1 \right] = \frac{\left(e^{2\pi(1-ik)} - 1 \right)}{2\pi(1-ik)}$$

$$e^{2\pi - 2\pi i k} = \frac{e^{2\pi}}{e^{2\pi i k}} = e^{2\pi}$$

$$e^{2\pi i k} = \underbrace{\cos(2\pi k)}_{1} + i \underbrace{\sin(2\pi k)}_{0}$$

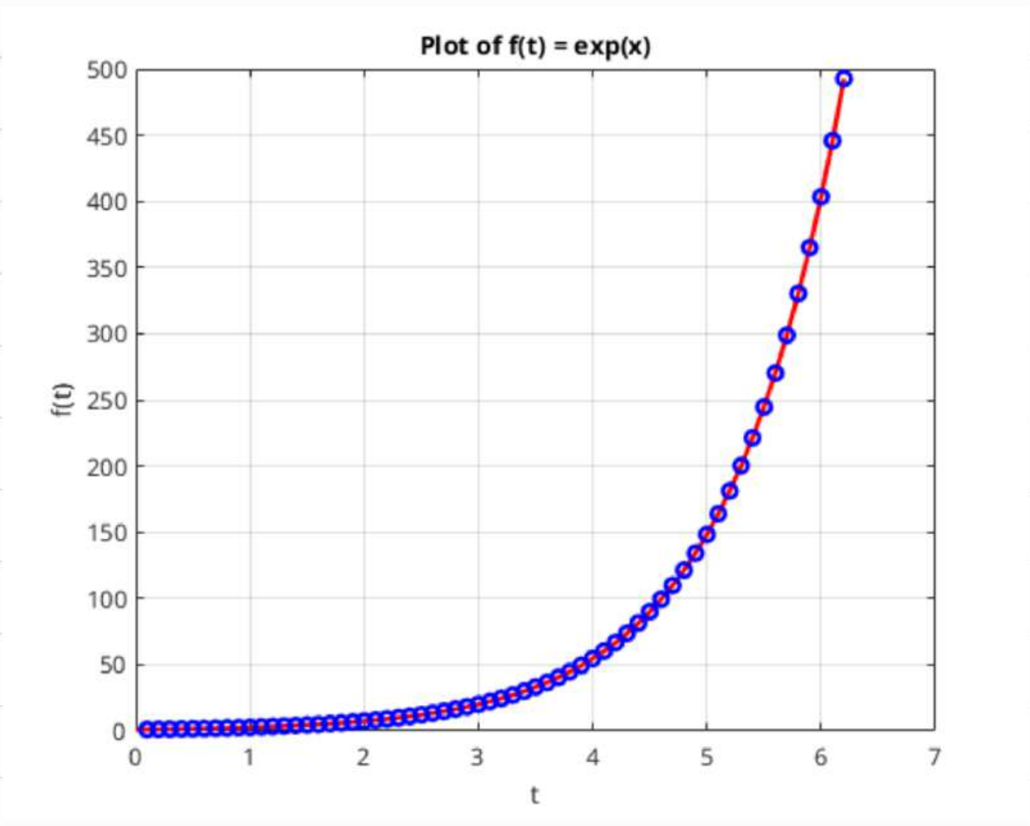
$$X_k = \frac{1}{2\pi} \cdot \frac{(e^{2\pi} - 1)}{(1-ik)} \qquad x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi} \cdot \frac{(e^{2\pi} - 1)}{(1-ik)} \cdot e^{i2\pi k t/T}$$

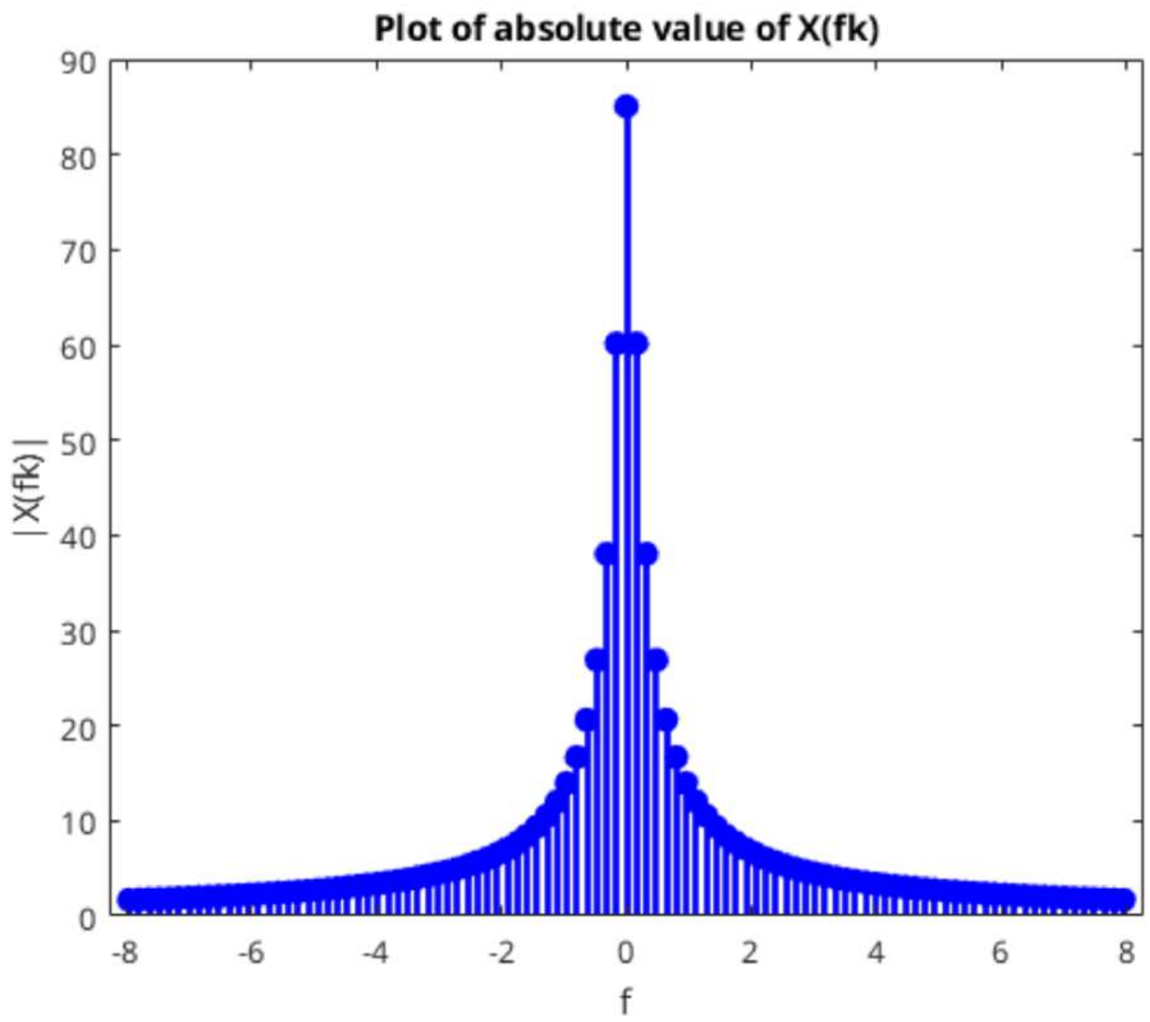
$T = 2\pi$ $$x(t) = \frac{1}{2\pi} \cdot \sum_{k=-\infty}^{\infty} \frac{(e^{2\pi} - 1)}{(1-ik)} \cdot e^{ikt}$$

$$|X_k| = \left| \frac{(e^{2\pi}-1)}{2\pi(i-ik)} \right| = \frac{|(e^{2\pi}-1)|}{|2\pi(i-k)|}$$

$$= \frac{|e^{2\pi}-1|}{|2\pi i - 2\pi k|} = \frac{e^{2\pi}-1}{\sqrt{(2\pi k)^2 + (2\pi)^2}} = \frac{e^{2\pi}-1}{\sqrt{(2\pi)^2(k^2+1)}}$$

$$= \frac{e^{2\pi}-1}{2\pi \cdot \sqrt{1+k^2}}$$




Plot of absolute value of X(fk)
|X(fk)|
f
90
80
70
60
50
40
30
20
10
0
-8
-6
-4
-2
0
2
4
6
8

Questão 1.3

$$f(t) = e^{2\cos(t)} \cdot \cos(2\sin(t))$$

1- Provar que é uma função periódica

Se $f(t) = f(t+T)$ para todo t pertencente ao domínio de f, então $f(t)$ é uma função periódica.

$$e^{2\cos(t)} \cdot \cos(2\sin(t)) = e^{2\cos(t+T)} \cdot \cos(2\sin(t+T))$$

A- $\cos(t) = \cos(t+T)$
e B- $\cos(2\sin(t)) = \cos(2\sin(t+T))$

A- $\cos(t) = \cos(t+T)$

$\cos(t) = \cos(t) \cdot \cos(T) - \sin(t) \cdot \sin(T)$

Se $T = 2\pi$, $\cos(2\pi) = 1$, $\sin(2\pi) = 0$

$\cos(t) = \cos(t) \cdot 1 = \cos(t)$

B- $\cos(2\sin(t)) = \cos(2\sin(t+T))$

$\cos^{-1}\cos(2\sin(t)) = \cos^{-1}\cos(2\sin(t+T))$

$2\sin(t) = 2\sin(t+T)$

$\sin(t) = \sin(t+T)$

$\sin(t) = \sin(t) \cdot \cos(T) + \sin(T) \cdot \cos(t)$

$T = 2\pi \quad \cos(2\pi) = 1$ e $\sin(2\pi) = 0$

Logo $\sin(t) = \sin(t)$

Como

A- $\cos(t) = \cos(t+T)$

e B- $\cos(2\sin(t)) = \cos(2\sin(t+T))$

São válidas para um mesmo período $T = 2\pi$

A relação abaixo é verdadeira

$$e^{2\cos(t)} \cdot \cos(2\sin(t)) = e^{2\cos(t+T)} \cdot \cos(2\sin(t+T))$$

e a função é periódica.

2- Provar que a expansão em série de Fourier

é igual a $1 + \sum_{n=1}^{\infty} \frac{2^n}{n!} \cos(nt)$

Vamos utilizar o intervalo de 0 a 2π.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi k t/T}$$

$$X_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi k t/T}\, dt$$

$T = 2\pi$

$$X_k = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot \cos(2\sin(t)) \cdot e^{-i\cancel{2}\pi k t/2\pi}\, dt$$

$$X_k = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot \cos(2\sin(t)) \cdot e^{-ikt}\, dt$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\cos(2\sin(t)) = \frac{e^{i\cdot 2\sin(t)} + e^{-i2\sin(t)}}{2}$$

$$\frac{1}{4\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot e^{-ikt} \cdot \left(e^{i2\sin(t)} + e^{-i2\sin(t)} \right) dt$$

$$\cos(t) = \frac{e^{it} + e^{-it}}{2} \qquad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

$$e^{2\cos(t)} = e^{\cancel{2}\cdot\frac{(e^{it}+e^{-it})}{\cancel{2}}} = e^{e^{it}+e^{-it}}$$

$$e^{i2\sin(t)} = e^{i\cancel{2}\cdot\frac{(e^{it}-e^{-it})}{\cancel{2i}}} = e^{e^{it}-e^{-it}}$$

$$e^{-i2\sin(t)} = e^{-i\cancel{2}\frac{(e^{it}-e^{-it})}{\cancel{2i}}} = e^{e^{-it}-e^{it}}$$

$$e^{it} = A \qquad e^{-it} = B$$

$$\frac{1}{4\pi}\cdot\int_0^{2\pi} e^{A+B}\cdot e^{-ikt}\cdot\left(e^{A-B}+e^{B-A}\right)dt$$

$$\frac{1}{4\pi}\left\{\int_0^{2\pi} e^{A+B+A-B-ikt}\,dt + \int_0^{2\pi} e^{A+B+B-A-ikt}\,dt\right\}$$

$$\frac{1}{4\pi}\cdot\left\{\int_0^{2\pi} e^{2A-ikt}\,dt + \int_0^{2\pi} e^{2B-ikt}\,dt\right\}$$

$$e^{-ikt}\cdot\left(e^{2e^{it}} + e^{2e^{-it}}\right)$$

$$e^{it} = \cos(t) + i\sin(t)$$

$$e^{-it} = \cos(t) - i\sin(t)$$

$$e^{2(\cos(t) + i\sin(t))} + e^{2(\cos(t) - i\sin(t))}$$

$$e^{2\cos(t)} \left(e^{2i\sin(t)} + e^{-2i\sin(t)} \right)$$

$$e^{2i\sin(t)} = \cos(2\sin(t)) + i\cancel{\sin}(2\sin(t))$$

$$e^{-2i\sin(t)} = \cos(2\sin(t)) - i\cancel{\sin}(2\sin(t))$$

$$= 2\cos(2\sin(t))$$

Segunda tentativa

Série de Taylor de uma exponencial

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$f(t) = e^{2\cos(t)} \cdot \cos(2\sin(t))$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi kt/T}$$

$$X_k = \frac{1}{T}\int_0^T x(t) \cdot e^{-i2\pi kt/T} \, dt$$

$$\cos(2\sin(t)) = \frac{e^{i\cdot(2\sin(t))} + e^{-i(2\sin(t))}}{2}$$

$$f(t) = \frac{e^{2\cos(t) + 2i\sin(t)} + e^{2\cos(t) - 2i\sin(t)}}{2}$$

$$X_k = \frac{1}{T}\int_0^T x(t) \cdot e^{-i2\pi kt/T} \, dt$$

$T = 2\pi$

$$x_K = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2\cos(t) + 2i\sin(t) - ikt}\, dt + \int_0^{2\pi} e^{2\cos(t) - 2i\sin(t) - ikt}\, dt$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sum_{n=0}^{\infty} \frac{(2\cos(t) + 2i\sin(t) - ikt)^n}{n!} = \sum_{n=0}^{\infty} \frac{(2e^{it} - ikt)^n}{n!}$$

$$\cos(t) + i\sin(t) = e^{it}$$

$$= \sum_{n=0}^{\infty} \frac{(2e^{-it} - ikt)^n}{n!}$$

$$\sum_{n=0}^{\infty} \int_0^{2\pi} \frac{(2e^{it} - ikt)^n + (2e^{-it} - ikt)^n}{n!}\, dt$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_0^{2\pi} (2e^{+it} - ikt)^n + (2e^{-it} - ikt)^n \, dt$$

$\Downarrow$

L

$$L \cdot \left\{ \int_0^{2\pi} (2e^{it} - ikt)^n \, dt + \int_0^{2\pi} (2e^{-it} - ikt)^n \, dt \right.$$

terceira Tentativa

$$f(t) = e^{2\cos(t)} \cdot \cos(2\sin(t))$$

$$\cos(2\sin(t)) = \frac{e^{i2\sin(t)} + e^{-i2\sin(t)}}{2}$$

$$f(t) = e^{2\cos(t)} \cdot \left(\frac{e^{i2\sin(t)} + e^{-i2\sin(t)}}{2} \right)$$

$$\cos(t) = \frac{e^{it} + e^{-it}}{2} \qquad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

$$f(t) = e^{(e^{it} + e^{-it})} \cdot \left(\frac{e^{(e^{it} - e^{-it})} + e^{(e^{-it} - e^{it})}}{2} \right)$$

$$f(t) = \frac{\left(e^{2e^{it}} + e^{2e^{-it}} \right)}{2}$$

Série de Taylor de uma exponencial

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$p(t) = e^{2e^{it}} \qquad g(t) = e^{2e^{-it}}$$

$$f(t) = \frac{p(t) + g(t)}{2}$$

$$p(t) = e^{x}, \quad x = 2e^{it}; \quad g(t) = e^{y}, \quad y = e^{2e^{-it}}$$

$$p(t) = \sum_{n=0}^{\infty} \frac{(2e^{it})^n}{n!} \qquad g(t) = \sum_{n=0}^{\infty} \frac{(2e^{-it})^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(2e^{it})^{-n}}{n!}$$

$$=$$

$$f(t) = \frac{1}{2} \cdot \left\{ \sum_{n=0}^{\infty} \frac{(2e^{it})^n}{n!} + \sum_{n=0}^{n=\infty} \frac{(2e^{in})^{-n}}{n!} \right\}$$

$$f(t) = \frac{1}{2} \cdot \left\{ \sum_{n=0}^{\infty} \frac{(2e^{it})^n + (2\overset{+}{e}^{in})^n}{n!} \right\}$$

$n=0$

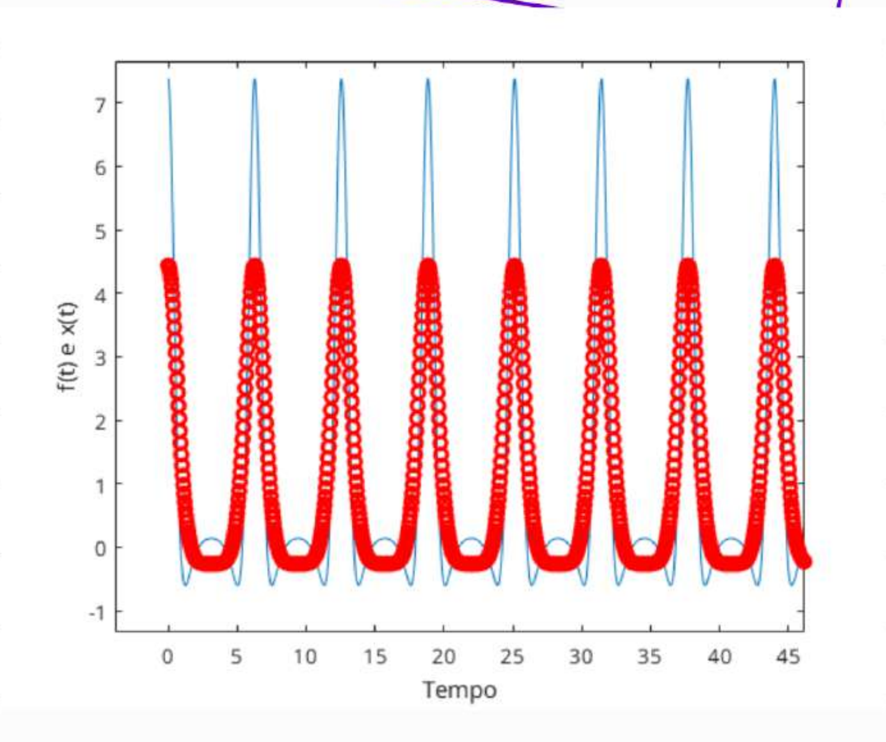
$$\frac{1}{2}\cdot\left\{ \frac{1+1}{1} \right\} = 1$$

$$f(t) = 1 + \frac{1}{2}\cdot\sum_{n=1}^{\infty} \frac{2e^{itn} + 2e^{-itn}}{n!}$$

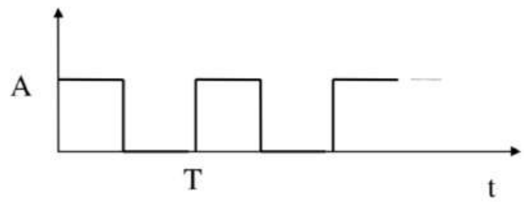
$$f(t) = 1 + \sum_{n=1}^{\infty} \frac{e^{itn} + e^{-itn}}{n!}$$

$$\cos(nt) = \frac{e^{itn} + e^{-itn}}{2}$$

$$\boxed{f(t) = 1 + \sum_{n=1}^{\infty} \frac{2\cdot\cos(nt)}{n!}}$$



1.4 Calcular os coeficientes da série de Fourier complexa para uma onda quadrada de período T e amplitude pico-a-pico A mostrada na figura abaixo:



$$x(t) = \sum_{k=0}^{\infty} X_k \cdot e^{i2\pi kt/T} \qquad X_k = \frac{1}{T}\int_0^T x(t) \cdot e^{-i2\pi kt/T}$$

$$x(t) = \begin{cases} A, & T.P \leq t \leq T.P + \frac{T}{2} \\ 0, & T.P + \frac{T}{2} < t < T.(P+1) \end{cases} \qquad P \in \mathbb{N}$$

Com $P = 0$

$$x(t) = \begin{cases} A, & 0 \leq t \leq T/2 \\ 0, & T/2 < t < T \end{cases}$$

$$X_k = \frac{1}{T}\int_0^{T/2} A \cdot e^{-i2\pi kt/T} dt = \frac{A}{T} \cdot T \left[\frac{e^{-i2\pi kt/T}}{-i2\pi k} \right]_0^{T/2}$$

$$= \frac{-A}{i2\pi k} \left[e^{-i2\pi k T/2T} - 1 \right] = \frac{-A}{i2\pi k} \cdot \left[e^{-i\pi k} - 1 \right]$$

$$e^{-i\pi k} = \underbrace{\cos(\pi k)}_{\substack{-1 \text{ para } k \text{ ímpar} \\ 1 \text{ para } k \text{ par}}} - i \underbrace{\sin(\pi k)}_{0}$$

$$X_k = \begin{cases} \frac{A}{i\pi k}, & k \text{ ímpar} \rightarrow \frac{-Ai}{\pi k}, \ |X_k| = \frac{|A|}{\pi |k|} \\ 0, & k \text{ par} \end{cases}$$

$$x(t) = \sum_{k=1,3,5,7,\dots}^{\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T} + \sum_{k=-1,-3,-5}^{-\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T}$$

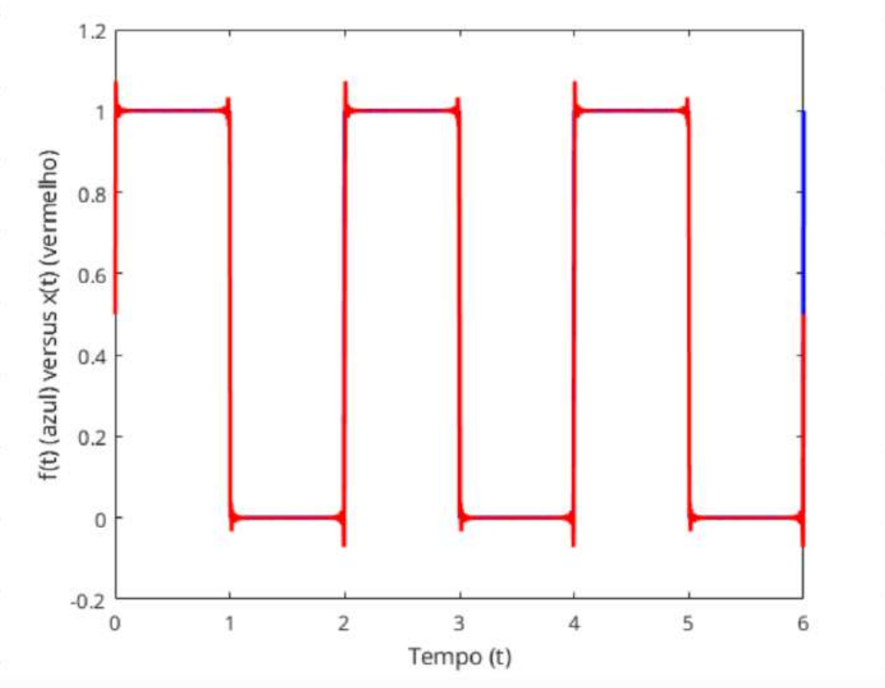
$$x(t) = \sum_{k=1,3,5,7,\dots}^{\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T} - \frac{A}{i\pi k} \cdot e^{-i2\pi k t/T}$$

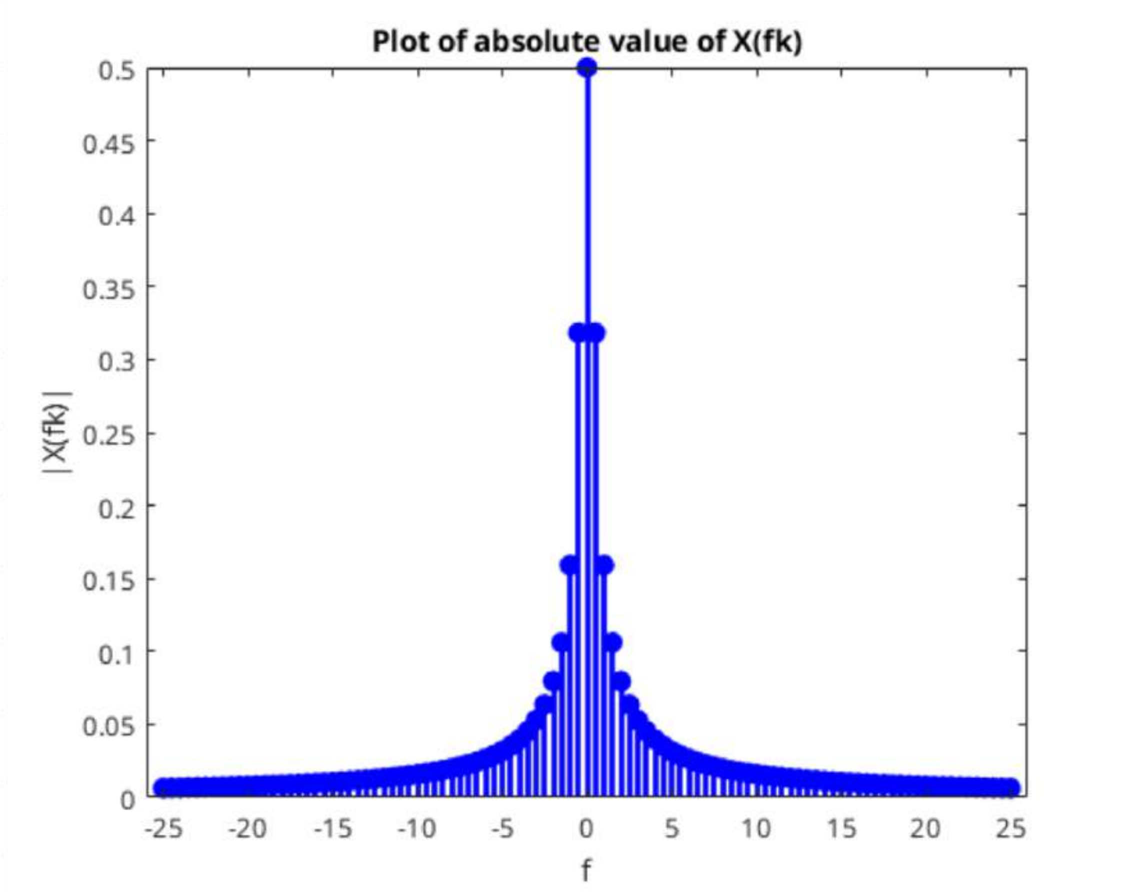
$$x(t) = \frac{A}{i\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{1}{k} \cdot \left(e^{i2\pi k t/T} - e^{-i2\pi k t/T} \right)$$

$$x(t) = \frac{A}{\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{2}{k} \cdot \left(\sin\left(2\pi k t/T\right)\right)$$

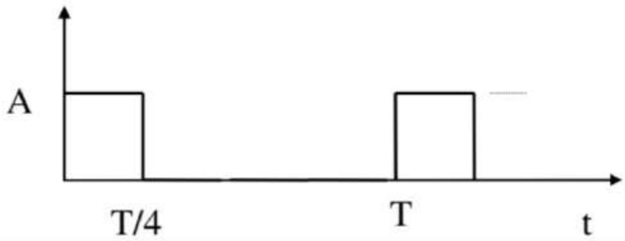
$$X_0 = \frac{1}{T} \cdot \int_0^{T/2} A \cdot e^{-i2\pi \cdot 0 \cdot t/T} = \frac{A \cdot T}{T \cdot 2} = \frac{A}{2}$$

$$x(t) = \frac{A}{2} + \frac{A}{\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{2}{k} \cdot \left(\sin\left(2\pi k t/T\right)\right)$$




Plot of absolute value of X(fk)
|X(fk)|
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-25
-20
-15
-10
-5
0
5
10
15
20
25
f

1.5 Calcular os coeficientes da série de Fourier complexa para a onda de período T e amplitude pico-a-pico A mostrada na figura abaixo:



$$x(t) = \begin{cases} A, & T.\rho \le t \le T.\rho + T/4 \\ 0, & T.\rho + T/4 < t < T.(\rho+1) \end{cases} \qquad \rho \in \mathbb{N}$$

$\rho = 0$

$$x(t) = \begin{cases} A, & 0 \le t \le T/4 \\ 0, & T/4 < t < T \end{cases}$$

$$x_k = \frac{1}{T}\int_0^{T/4} A.\, e^{-i2\pi k t/T}\, dt = \frac{A}{T} \cdot \left[\frac{e^{-i2\pi k t/T}}{-i2\pi k/T} \right]_0^{T/4}$$

$$\frac{A}{T} \cdot \frac{T}{-i2\pi k} \cdot \left[e^{-i2\pi k T/4T} - 1 \right] = \frac{-A}{i2\pi k} \cdot \left[e^{-i\pi k/2} - 1 \right]$$

$$e^{-i\pi k/2} = \cos(\pi k/2) - i\sin(\pi k/2)$$

$k = 0 \quad = 1$

$k = 1 \quad = -i$

$k = 2 \quad = -1$

$k = 3 \quad = i$

$$x_0 = \frac{A}{T} \cdot \int_0^{T/4} 1\, dt = \frac{A}{T} \cdot \frac{T}{4} = \frac{A}{4}$$

$$x_k = \begin{cases} \frac{A}{4}, & k = 0 \\ 0, & k \% 4 = 0, k != 0 \\ \frac{A}{i2\pi k}(i+1), & k = 1 \\ \frac{A}{i\pi k}, & k = 2 \\ \frac{A}{i2\pi k}(1-i), & k = 3 \end{cases}$$

0, 1, 2, 3,
4, 5, 6, 7
8, 9, 10, 11

Intervalo se repete a cada 4.

$$X_k = \begin{cases} \frac{A}{4}, & k=0 \\ 0, & 4k, k \neq 0 \\ \frac{A}{i2\pi k}(i+1), & 4k+1 \\ \frac{A}{i\pi k}, & 4k+2 \\ \frac{A}{i2\pi k}(1-i), & 4k+3 \end{cases} \quad k = 0, 1, 2, 3, \ldots$$

$\sqrt{1^2+1^2} = \sqrt{2}$

$1 \cdot i$

$k \% 4 = 1$

$$|X_k| = \frac{|A(i+1)|}{|i2\pi k|} = \frac{|A| \cdot |i+1|}{|i| \cdot |2\pi k|}$$

$$\frac{|A| \cdot \sqrt{2}}{2\pi |k|}$$

$k \% 4 = 2$

$$|X_k| = \frac{|A|}{|i2\pi k|} = \frac{|A|}{2\pi \cdot |k|}$$

$K \% 4 = 3$

$$|X_K| = \frac{|A| \cdot |i - 1|}{|i\, 2\pi K|} = \frac{|A| \cdot \sqrt{2}}{2\pi \cdot |K|}$$

$\to \sqrt{|1| + |-1|^2} = \sqrt{2}$

$$|X_K| = \begin{cases} 0, & K \% 4 == 0 \\ \dfrac{|A| \cdot \sqrt{2}}{2\pi \cdot |K|}, & K \% 4 = 1 \vee K \% 4 = 3 \\ \dfrac{|A|}{2\pi |K|}, & K \% 4 = 2 \end{cases}$$

$$\frac{-A}{i2\pi k} \cdot \left[e^{-i\pi k/2} - 1 \right] = \frac{A}{i2\pi k} \cdot \left(1 - e^{-i\pi k/2} \right)$$

$$e^{-i\pi k/2} = \cos(\pi k/2) - i \sin(\pi k/2)$$

k par

	$k \% 4 = 0$	1	$x_k = 0$
	$k \% 4 = 1$	$-i$	$x_k = A/(i2\pi k) \cdot (1+i)$
	$k \% 4 = 2$	-1	$x_k = A / i\pi k$
	$k \% 4 = 3$	i	$x_k = (A/2i\pi k) \cdot (1-i)$

$$\frac{A \cdot e^{-i\pi k/4}}{\pi k} \cdot \left(\frac{e^{i\pi k/4} - e^{-i\pi k/4}}{2i} \right)$$

$$\frac{A \cdot e^{-i\pi k/4}}{\pi k} \cdot \left(\sin(\pi k/4) \right)$$

Por Trigonometrica

$$x(t) = \frac{a_0}{2} + \sum_{K=1}^{\infty} \left[a_K \cos\left(K \frac{2\pi}{T} \cdot t\right) + b_K \cdot \sin\left(K \frac{2\pi}{T} \cdot t\right) \right]$$

$$\frac{A}{i2\pi K} \cdot \left[1 - e^{-i\pi K/2}\right] = X_K$$

$$a_K = X_K + X_{-K} \qquad b_K = j\left(X_K - X_{-K}\right)$$

$$a_K = \frac{A}{i2\pi K}\left[1 - e^{-i\pi K/2}\right] + \frac{A}{-i2\pi K}\left[1 - e^{i\pi K/2}\right]$$

$$a_K = \frac{A}{i2\pi K}\left[1 - e^{-i\pi K/2} - 1 + e^{i\pi K/2}\right]$$

$$a_K = \frac{A}{i2\pi K} \cdot \left(e^{i\pi K/2} - e^{-i\pi K/2} \right]$$

$$a_K = \frac{A}{\pi K} \cdot \sin\left(\pi K/2\right)$$

$$b_K = \frac{Ai}{i2\pi K}\left[1 - e^{-i\pi K/2} + 1 - e^{i\pi K/2}\right]$$

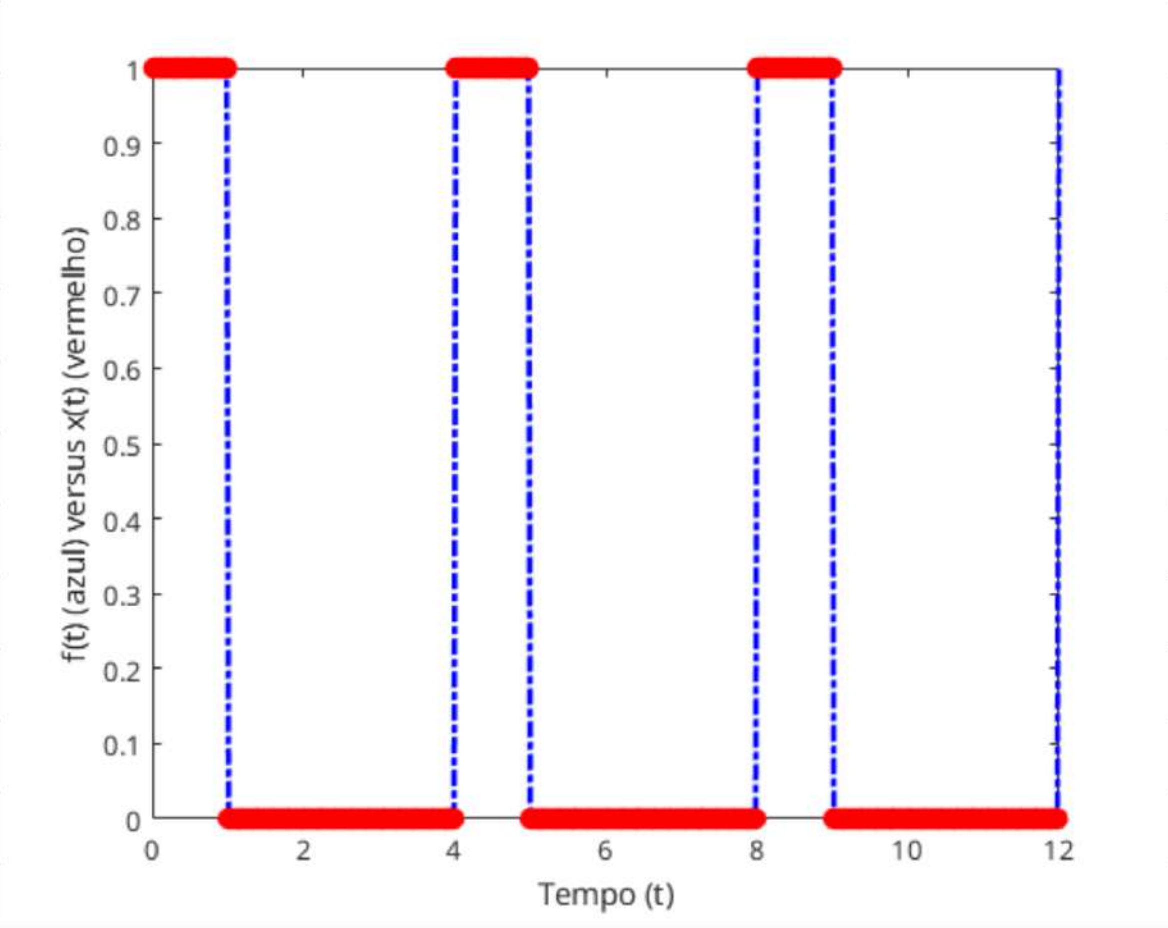
$$b_k = \frac{A}{\pi k} + \frac{-A}{2\pi k}\left[e^{i\pi k/2} + e^{-i\pi k/2}\right]$$

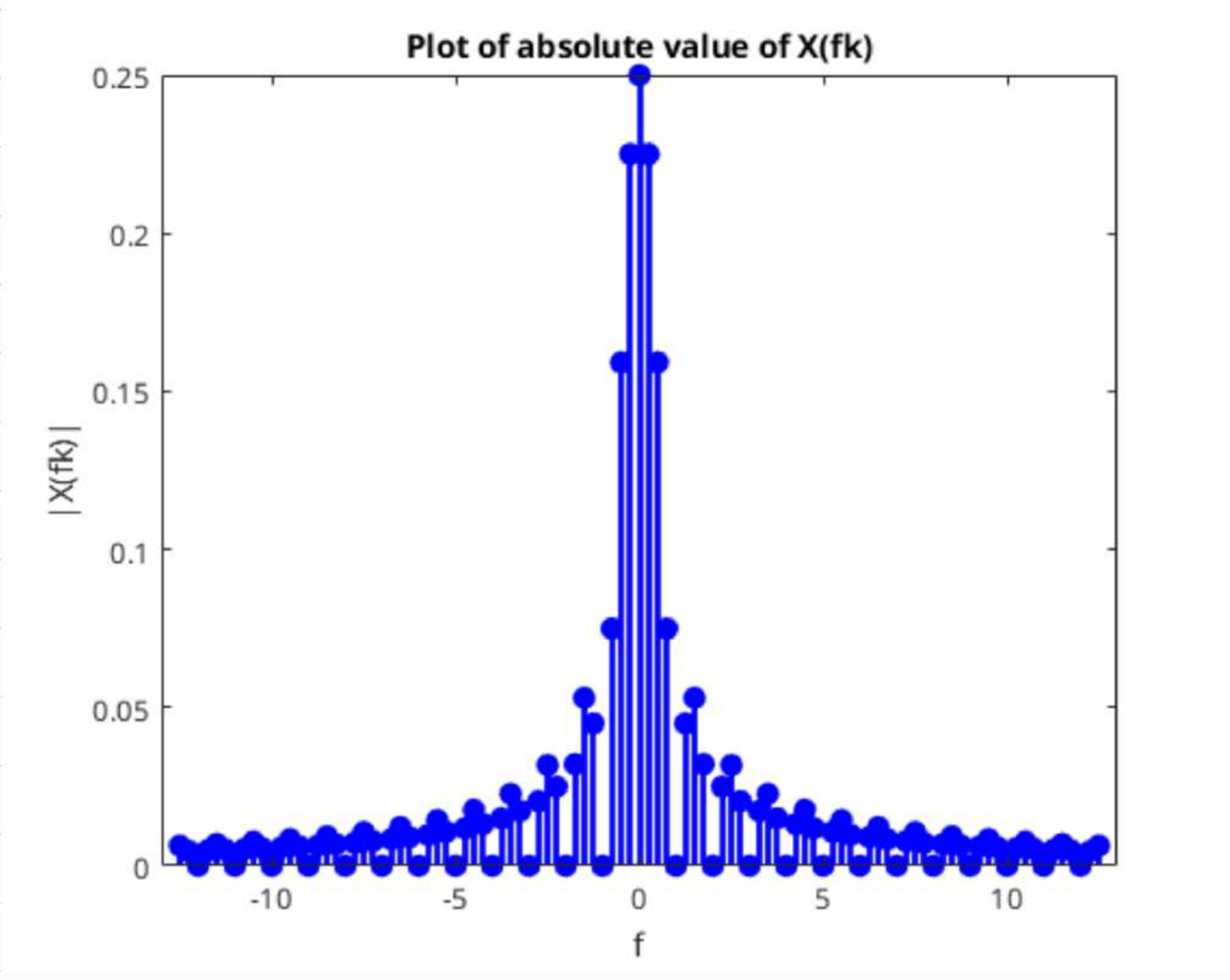
$$b_k = \frac{A}{\pi k} - \frac{A}{\pi k} \cdot \cos(\pi k/2)$$

$$b_k = \frac{A}{\pi k}\left(1 - \cos(\pi k/2)\right)$$

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty}\left[a_k \cos\left(k \frac{2\pi}{T} \cdot t\right) + b_k \cdot \sin\left(k \frac{2\pi}{T} \cdot t\right)\right]$$

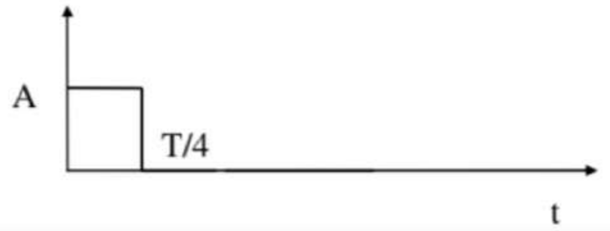
$$x(t) = \frac{A}{4} + \sum_{k=1}^{\infty} \frac{A}{\pi k} \cdot \left[\sin\left(\frac{\pi k}{2}\right) \cdot \cos\left(\frac{k2\pi}{T} \cdot t\right) + \left(1 - \cos(\pi k/2)\right) \cdot \sin\left(k \frac{2\pi}{T} \cdot t\right)\right]$$


f(t) (azul) versus x(t) (vermelho)
Tempo (t)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
2
4
6
8
10
12


Plot of absolute value of X(fk)
|X(fk)|
f
0
0.05
0.1
0.15
0.2
0.25
-10
-5
5
10

Questão 1.6

1.6 Calcular a transformada (integral) de Fourier do pulso retangular abaixo. Se possível, traçar os resultados dos itens anteriores usando o programa MATLAB (traçar somente os valores absolutos).



$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{i2\pi f t}\, df$$

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i2\pi f t}\, dt$$

$$x(t) = \begin{cases} A, & 0 \leq t \leq T/4 \\ 0, & t < 0 \vee t > T/4 \end{cases}$$

$$X(f) = \int_{0}^{T/4} A \cdot e^{-i2\pi f t}\, dt = A \cdot \left[\frac{e^{-i2\pi f t}}{-i2\pi f} \right]_{0}^{T/4}$$

$$= \frac{-A}{i2\pi f} \cdot \left[e^{-i2\pi f \cdot T/4} - 1 \right] \qquad f = \frac{1}{T}$$

$$= \frac{-A}{i2\pi f} \cdot \left[e^{-i\pi \frac{1}{T} \cdot \frac{T}{2}} - 1 \right] = \frac{-A}{i2\pi f} \left[e^{-i\pi/2} - 1 \right]$$

$$= \frac{A}{\pi f} \cdot e^{-i\pi/4} \left[\frac{e^{i\pi/4} - e^{-i\pi/4}}{2i} \right] = \frac{A}{\pi f} \cdot e^{-i\pi/4} \cdot \sin\left(\frac{\pi}{4}\right)$$

$$= \frac{A}{4f} \cdot e^{-i\pi/4} \cdot \frac{\sin(\pi/4)}{\pi} \cdot 4 = \frac{A}{4f} \cdot e^{-i\pi/4} \cdot \text{sinc}(\pi/4)$$

$$= \frac{A}{4f} \cdot \text{sinc}(\pi/4) \cdot e^{-i\pi/4}$$

$$|X(f)| = \sqrt{\underbrace{re}_{\text{parte real}}^2 + \underbrace{im}_{\text{parte imaginaria}}^2}$$

$$X(f) = \frac{A}{4f} \cdot \text{sinc}(\pi/4) \cdot \left(\cos(\pi/4) - i \sin\left(\frac{\pi}{4}\right)\right)$$

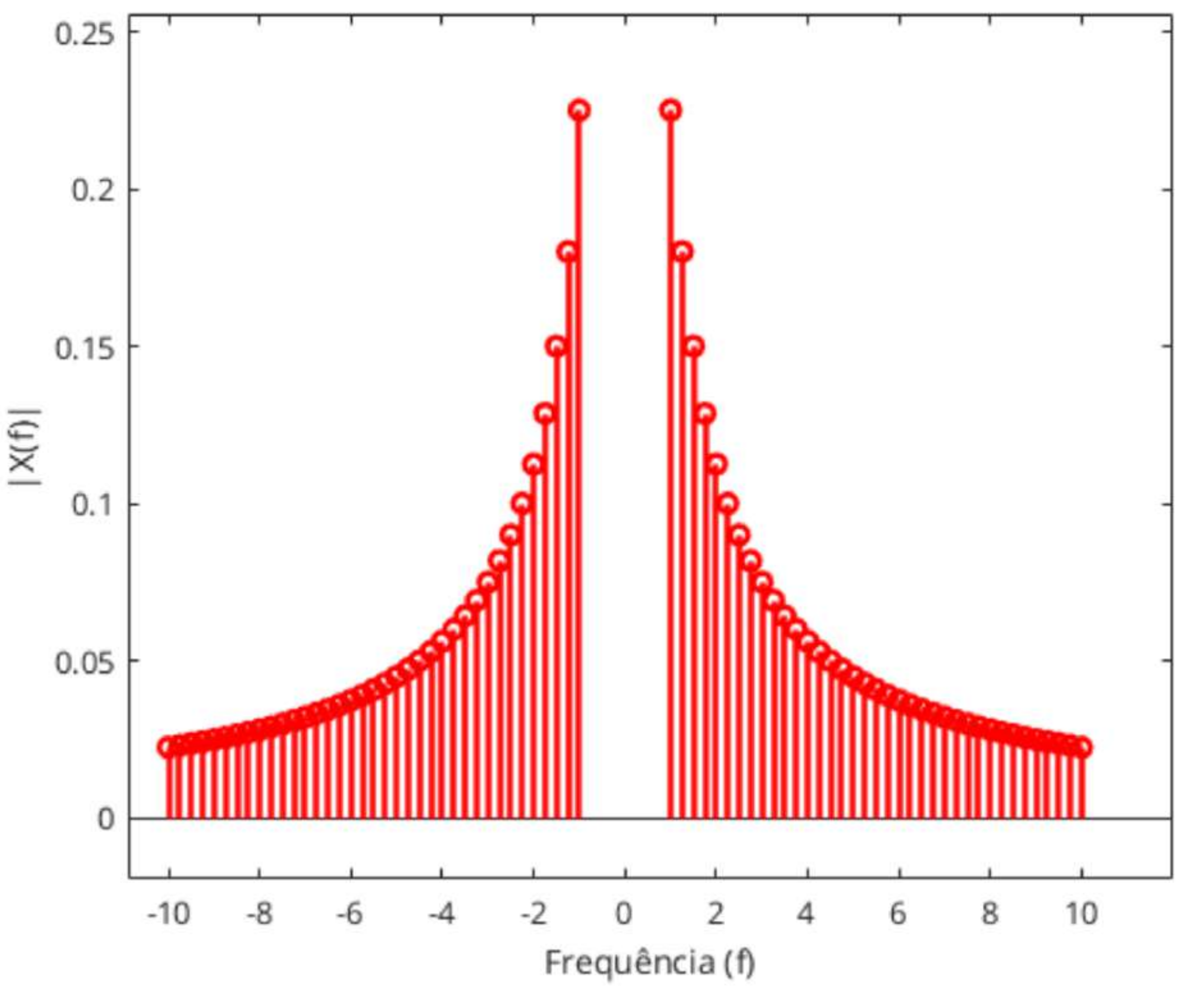
$$|X(f)| = \sqrt{\left(\frac{A}{4f} \cdot \text{sinc}(\pi/4)\right)^2} \cdot \sqrt{\cos(\pi/4)^2 - \sin\left(\frac{\pi}{4}\right)^2}$$

$$= \frac{A}{|4f|} \cdot |\text{sinc}(\pi/4)| \cdot \sqrt{\cos(-\pi/4)^2 + \sin\left(-\frac{\pi}{4}\right)^2}$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

$$= \frac{A}{|4f|} \cdot |\operatorname{sinc}(\pi/4)| \cdot \sqrt{1}$$

$$|X(f)| = \frac{A}{|4f|} \cdot |\operatorname{sinc}(\pi/4)|$$



Questão 1.7

1.7 Provar o teorema da convolução na forma:

$x(t)h(t) \Leftrightarrow X(f) * H(f)$

Transformada de Fourier

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{i2\pi ft} \, df$$

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i2\pi ft} \, dt$$

Convolução

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) \, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) \, d\tau = h(t) * x(t)$$

$$x(t) \cdot h(t) \overset{F}{\Longleftrightarrow} X(f) * H(f)$$

$$X(f) * H(f) = \int_{-\infty}^{\infty} X(\tau) \cdot H(f-\tau) \, d\tau$$

$$\int_{-\infty}^{\infty} x(t) \cdot h(t) \cdot e^{-i2\pi f t} dt = \int_{-\infty}^{\infty} X(\tau) \cdot H(f-\tau) d\tau$$

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi (f-\tau) t} dt \right] \cdot X(\tau) \, d\tau$$

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi f t} \cdot e^{i2\pi \tau t} dt \right] \cdot X(\tau) \, d\tau$$

Usando o teorema de Fubini

$$\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi f t} \cdot \left[\int_{-\infty}^{\infty} X(\tau) \cdot e^{i2\pi \tau t} d\tau \right] dt$$

Relembrando

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{i2\pi f t} dt$$

Logo

é igual a $x(t)$

$$= \int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi f t} \cdot x(t)\, dt$$

$$= \int_{-\infty}^{\infty} x(t) \cdot h(t) \cdot e^{-i2\pi f t} \cdot dt$$

Provando que

$$x(t) \cdot h(t) \overset{F}{\Longleftrightarrow} X(f) * H(f)$$

Questão 1.8

1.8 Mostrar que os coeficientes da Série de Fourier podem ser aproximados pela expressão:

$$X_k = \frac{1}{N}\sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi kt/T}$$

$$X_k = \frac{1}{T} \cdot \int_{-T/2}^{T/2} x(t) \cdot e^{-i2\pi t/T}\, dt = \frac{1}{T} \cdot \int_{0}^{T} x(t) \cdot e^{-i2\pi t/T}\, dt$$

A soma de Riemann à esquerda é definida como

$$\sum_{i=0}^{n-1} f(x_i) \cdot (x_{i+1} - x_i)$$

Sendo $f(t)$ uma função definida no intervalo $[a, b]$.

A integral definida de Riemann é definida como o limite da soma de Riemann quando n tende ao infinito.

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} f(x_i) \cdot \underbrace{(x_{i+1} - x_i)}_{\Delta x} = \sum_{i=0}^{\infty} f(x_i) \cdot \underbrace{(x_{i+1} - x_i)}_{\Delta x}$$

$$= \int_a^b f(t)\, dt$$

Voltando a definição de Xk

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi t/T} dt$$

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi t/T} dt \qquad f(t) = x(t) \cdot e^{-i2\pi t/T}$$

Reescrevendo a integral como uma soma de Riemann com n tendendo ao limite

$$\sum_{i=0}^{\infty} f(x_i) \cdot \underbrace{(x_{i+1} - x_i)}_{\Delta x} = \sum_{n=0}^{\infty} f(t_n) \cdot \underbrace{(t_{n+1} - t_n)}_{\Delta t}$$

$$\Delta t = \frac{T - 0}{N} \qquad t_n = n \Delta t$$

$$x_n = x(t_n)$$

número de intervalos

$$\sum_{n=0}^{\infty} f(t_n) \cdot \Delta t = \lim_{N \to \infty} \sum_{n=0}^{N} f(t_n) \cdot \Delta t$$

$$\approx \sum_{n=0}^{N-1} f(t_n) \cdot \frac{T}{N} = \frac{T}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k t_n / T}$$

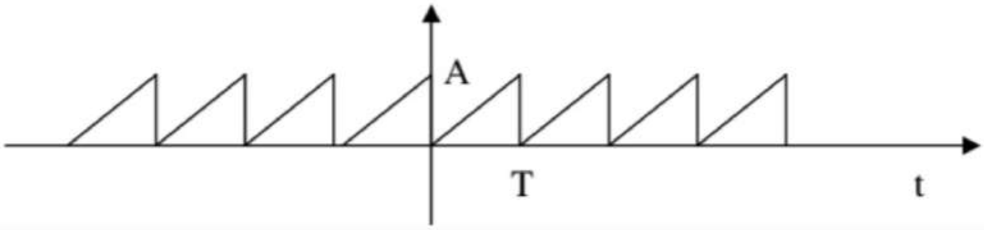
$$= \frac{T}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n T / NT} = \frac{T}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

$$X_k = \frac{1}{T} \cdot \int_0^T f(t)\,dt \approx \frac{1}{T} \cdot \frac{T}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

$$= \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

Questão 1.9

1.9 Calcular a série de Fourier de um sinal do tipo dente de serra usando a aproximação acima e comparar com a Série de Fourier analítica. Comparar os espectros graficamente. Usar A = 3 e T = 7



$$\hat{X}_k = \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n/N} \quad , x_n = x(t_n)$$

$$t_n = n \cdot \Delta t = n \cdot \frac{T}{N}$$

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi k t/T} \, dt$$

$$x(t) = \frac{A t}{T} \qquad x\left(t = n\frac{T}{N}\right) = A \cdot \frac{n T}{N \cdot T}$$

$$\hat{X}_k = \frac{A}{N^2} \cdot \sum_{n=0}^{N-1} n \cdot e^{-i2\pi k n/N} \qquad = \frac{A n}{N}$$

$$\boxed{x(t) = \sum_{k=-\infty}^{\infty} \frac{A}{N^2} \cdot \sum_{n=0}^{N-1} n \cdot e^{-j2\pi k n / N} \Big] \cdot e^{j2\pi k t/T}}$$

Agora Fourier Analítico

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi k t/T} \, dt$$

$$X_k = \frac{1}{T} \cdot \int_0^T \frac{At}{T} \cdot e^{-i2\pi k t/T} \, dt$$

$$X_k = \frac{A}{T^2} \cdot \underbrace{\int_0^T t \cdot e^{-i2\pi k t/T} \, dt}$$

$$\int u \, dv = u \cdot v - \int v \, du$$

$$u = t \quad du = dt \quad dv = e^{-i2\pi k t/T} \quad v = \frac{T \cdot e^{-i2\pi k t/T}}{-i2\pi k}$$

$$\left. \frac{T \cdot t \cdot e^{-i2\pi k t/T}}{-i2\pi k} \right]_0^T - \int_0^T \frac{T \cdot e^{-i2\pi k t/T}}{-i2\pi k} \, dt$$

$$\frac{-T^2 \cdot e^{-i2\pi k}}{i2\pi k} + \frac{T}{i2\pi k} \cdot \left[e^{-i2\pi k t/T} \right]_0^T$$

$$\frac{T}{i2\pi k}\left(-T.e^{-i2\pi k}+e^{-i2\pi k}-1\right]$$

$$e^{-i2\pi k} = \underbrace{\cos(2\pi k)}_{1} - i\underbrace{\sin(2\pi k)}_{0},\ k=0,1,2,\dots$$

$$\frac{T}{i2\pi k}\left(-T+1-1\right) = -\frac{T^2}{i2\pi k}$$

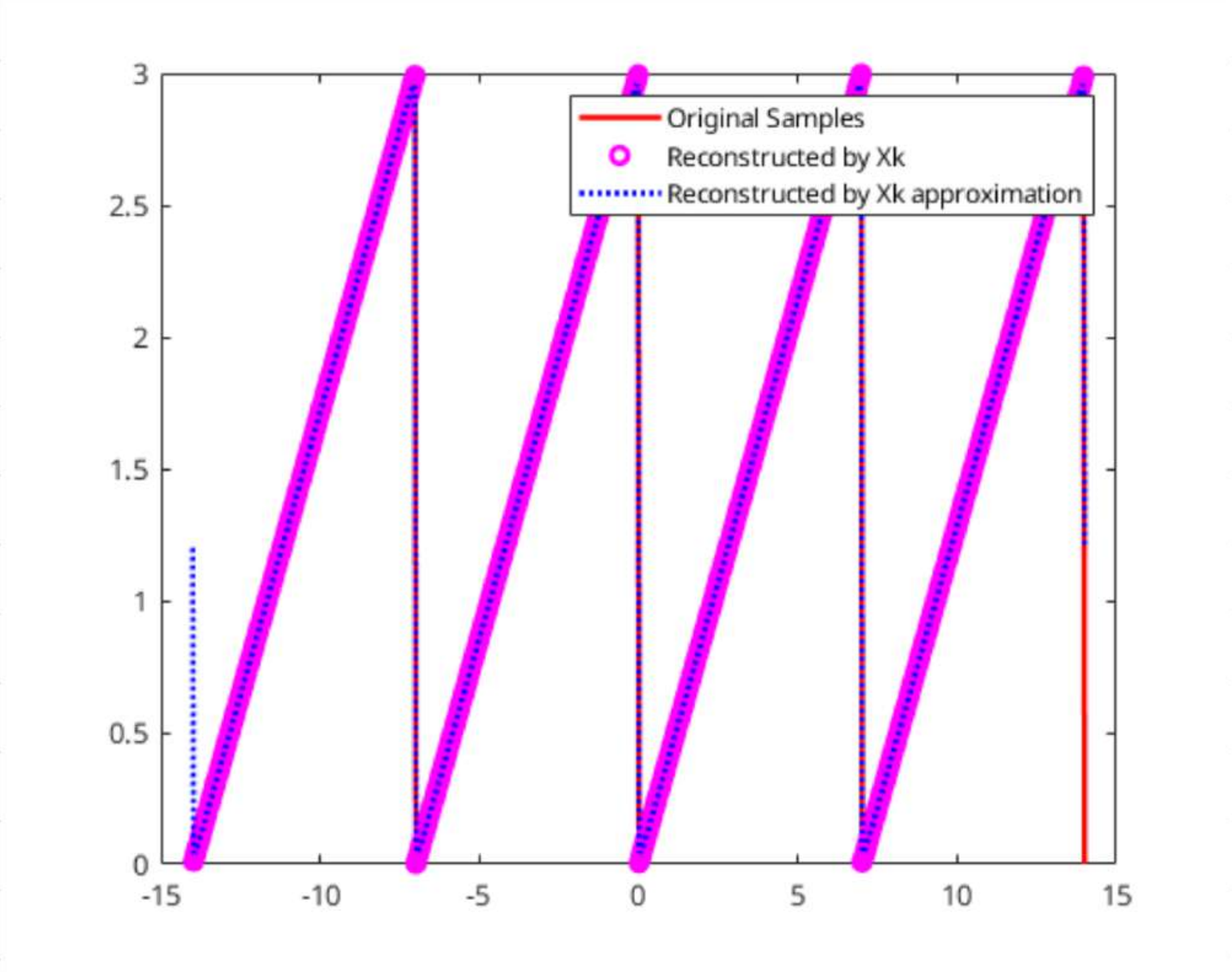
Voltando a equação original

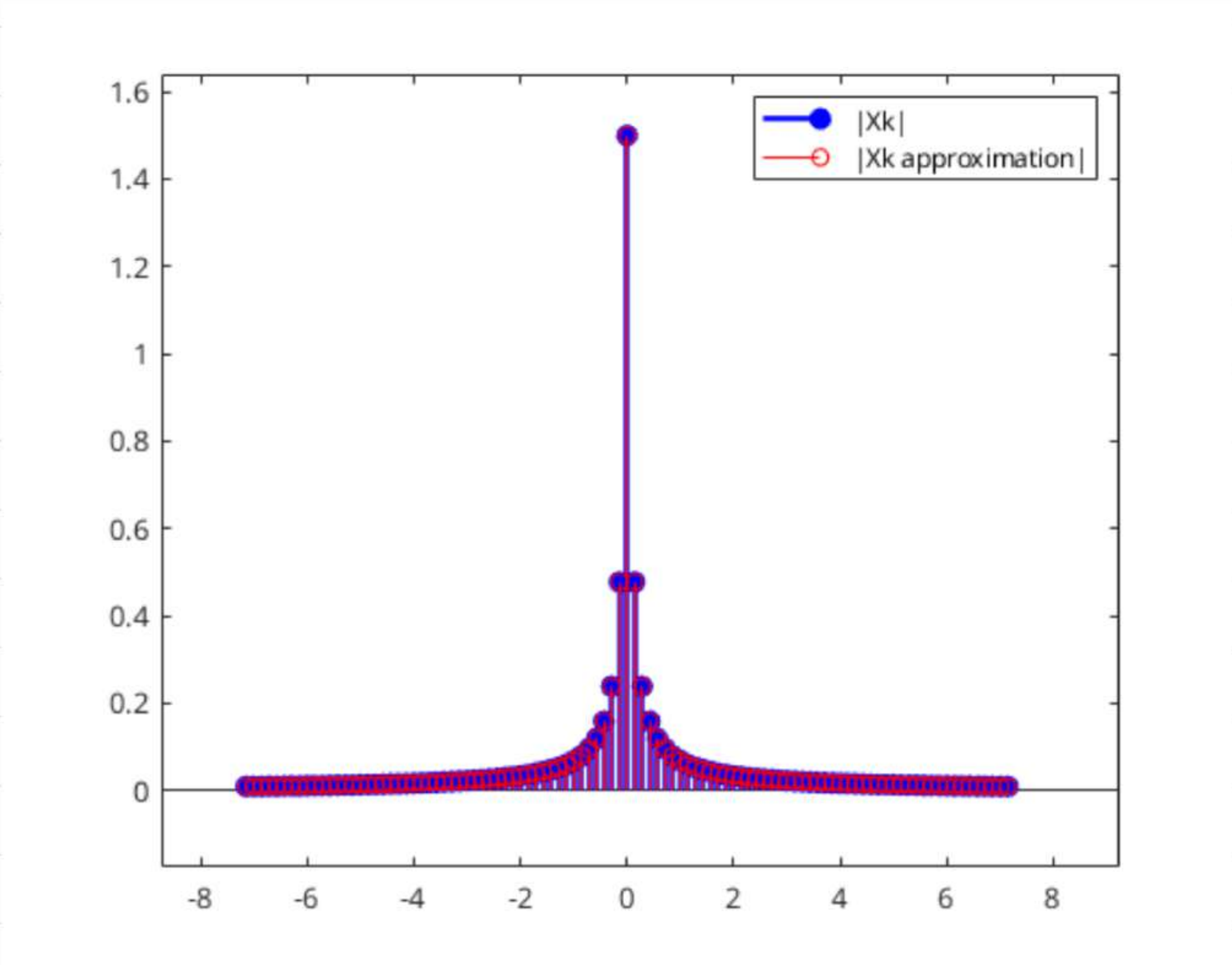
$$X_k = \frac{A}{T^2}\cdot\frac{-T^2}{i2\pi k} = \frac{Ai}{2\pi k}$$

$$X_0 = \frac{1}{T}\int_0^T \frac{A}{T}t\,dt = \frac{A}{T^2}\cdot\left[\frac{t^2}{2}\right]_0^T$$

$$= \frac{A}{2}$$

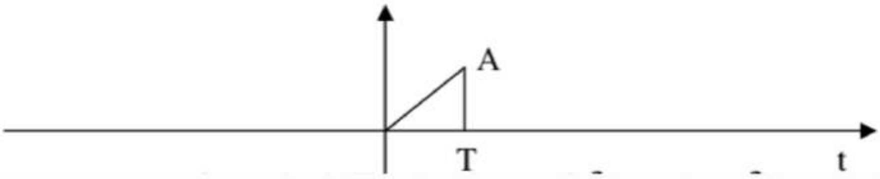
$$x(t) = \frac{A}{2} + \sum_{k=1}^{\infty}\frac{Ai}{2\pi k}\cdot e^{j2\pi kt/T} + \sum_{k=-\infty}^{-1}\frac{Ai}{2\pi k}\cdot e^{j2\pi kt/T}$$


Original Samples
Reconstructed by Xk
Reconstructed by Xk approximation
3
2.5
2
1.5
1
0.5
0
-15
-10
-5
0
5
10
15


|Xk|
|Xk approximation|
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-8
-6
-4
-2
0
2
4
6
8

Questão 10

1.10 Calcular a Transformada de Fourier de um pulso triangular (um período do sinal do item anterior) usando a aproximação da SF e a fórmula de Poisson.



$$\hat{X}_k = \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \, e^{-i 2\pi k n / N}$$

$$t_n = n \cdot \Delta t = n \cdot \frac{T}{N}$$

$$x(t) = \frac{A}{T} \cdot t$$

$$x(t)\Big|_{t=t_n} = \frac{A}{T} \cdot n \cdot \frac{T}{N} = \frac{An}{N}$$

Fórmula de Poisson:

$$\sum_{n=-\infty}^{\infty} x(t + nT) = \frac{1}{T} \cdot \sum_{n=-\infty}^{\infty} X(nf_0) \cdot e^{i 2\pi f_0 t}$$

1- Calcular $\hat{X}_K$

$$\hat{X}_K = \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi K n / N}$$

$$\hat{X}_K = \frac{1}{N} \cdot \sum_{n=0}^{N-1} \frac{A \cdot n}{N} \cdot e^{-i2\pi K n / N}$$

$$= \frac{A}{N^2} \cdot \sum_{n=0}^{N-1} n \cdot e^{-i2\pi K n / N} = \frac{A}{N^2} \cdot \sum_{n=1}^{N-1} n \cdot e^{-i2\pi K n / N}$$

2- Poisson

A transformada de Fourier de um sinal periódico pode ser dada por:

$$X(f) = \sum_{K=-\infty}^{\infty} X\left(\frac{K}{T}\right) \cdot \delta\left(f - \frac{K}{T}\right)$$

Os coeficientes $\hat{X}_K$ correspondem a amostras da transformada de Fourier $X(f)$ nos pontos $\frac{K}{T}$

3-

$$X(f) = \int_{-\infty}^{\infty} x(t) . e^{-i2\pi f t} dt$$

Fórmula de Poisson:

$$\sum_{n=-\infty}^{\infty} x(t + nT) = \frac{1}{T} . \sum_{n=-\infty}^{\infty} X(nf_0) . e^{i2\pi f_0 t}$$

$$X_n = \frac{A}{N^2} . \sum_{n=1}^{N-1} n . e^{-i2\pi k n / N}$$

$$X(f) = \int_{-\infty}^{\infty} \frac{1}{T} , \frac{A}{N^2} . \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{k=\infty} n . e^{-i2\pi k n / N} . e^{i2\pi f_0 t} dt$$

Com $T \to \infty$ e $f_0 = \frac{1}{T}$

Questão 11

1.11 Demonstre que a transformada de Fourier de: $a+b\delta(t-t_0)+c\delta'(t-t_1)$ é $2\pi a\delta(\omega)+be^{-i\omega t_0}+ic\omega t_1 e^{-i\omega t_1}$

$$a + b\delta(t-t_0) + c\delta'(t-t_1)$$

$$\xrightarrow{F} 2\pi a\delta(\omega) + b e^{-i\omega t_0} + ic\omega . e^{-i\omega t_1}$$

$$= F(a) + F(b\delta(t-t_0)) + F(ic.\delta'(t-t_1))$$

$$= a.\underbrace{F(1)}_{①} + b.\underbrace{F(\delta(t-t_0))}_{②} + ic.\underbrace{F(\delta'(t-t_1))}_{③}$$

① $\delta(1) = 2\pi\delta(\omega)$

② $F[\delta(t-t_0)] = \int_{-\infty}^{\infty} \delta(t-t_0) . e^{-i2\pi f t} dt$

$$= e^{-i2\pi f t_0} = e^{-i\omega t_0}$$

③ $F[\delta'(t-t_1)]$

Usando a propriedade da diferenciação

$$F[\dot{x}(t)] = j2\pi f \cdot X(f)$$

$$F[\dot{\delta}(t-t_1)] = i2\pi f \cdot F[\delta(t-t_1)]$$

$$= i2\pi f \cdot e^{-i2\pi f t_1} = i\omega \cdot e^{-j\omega t_1}$$

Juntando ①, ② e ③ na equação original

$$a \cdot ① + b \cdot ② + iC \cdot ③$$

$$= \boxed{a \cdot 2\pi \cdot \delta(\omega) + b \cdot e^{i\omega t_0} + iC \cdot j\omega \cdot e^{-i\omega t_1}}$$

(12)

$$\ddot{x}(t) + 5\dot{x}(t) + 6x(t) = \delta(t) \qquad x(t) = \left(e^{-2t} + e^{-3t}\right)u(t)$$

$$F(x(t)) = X(f)$$
$$F(\dot{x}(t)) = j2\pi f X(f)$$
$$F(\ddot{x}(t)) = (j2\pi f)^2 \cdot X(f) = -(2\pi f)^2 \cdot X(f)$$
$$F(\delta(t)) = 1$$

$$-(2\pi f)^2 \cdot X(f) + 10 j\pi f X(f) + 6X(f) = 1$$

$$X(f)\left(-(2\pi f)^2 + 10j\pi f + 6\right) = 1$$

$$X(f) = \frac{1}{-(2\pi f)^2 + 10j\pi f + 6} = \frac{1}{2} \cdot \frac{1}{-2\pi^2 f^2 + 5j\pi f + 3}$$

Achando os polos de $X(f)$

$$2\pi^2 f - 5j\pi f - 3 = 0$$

$$\frac{-(-5j\pi) \pm \sqrt{(-5j\pi)^2 - 4 \cdot 2\pi^2 \cdot -3}}{4\pi^2}$$

$$\frac{5j\pi \pm \sqrt{-25\pi^2 + 24\pi^2}}{4\pi^2} = \frac{5j\pi \pm (j\pi)}{4\pi^2}$$

$$= \frac{5j \pm j}{4\pi} \quad \begin{cases} f_1 = \dfrac{j}{\pi} \\ f_2 = \dfrac{3j}{2\pi} \end{cases}$$

$$\left(f-\frac{j}{\pi}\right)\cdot\left(f-\frac{3j}{2\pi}\right)=\left(\frac{f\pi-j}{\pi}\right)\cdot\left(\frac{f2\pi-3j}{2\pi}\right)$$

$$\frac{f^2 2\pi^2-3j\pi f-2j\pi f+3}{2\pi^2}=\frac{2\pi^2 f^2-5j\pi f+3}{2\pi^2}$$

Voltando a $X(f)$

$$X(f)=\frac{1}{\left(f-\frac{j}{\pi}\right)\cdot\left(f-\frac{3j}{2\pi}\right)}\cdot\frac{1}{2\cdot 2\pi^2}$$

$$X(f)=\frac{1}{4\pi^2}\cdot\frac{1}{\left(f-\frac{j}{\pi}\right)\cdot\left(f-\frac{3j}{2\pi}\right)}=\frac{1}{4\pi^2}\cdot\frac{1}{(f-\alpha)\cdot\left(f-\frac{3\alpha}{2}\right)}$$

$\frac{j}{\pi}=\alpha$

Pelo princípio da linearidade

$$\rho\cdot x(t)\overset{F}{\Longrightarrow}\rho\cdot X(f)$$

$$\rho=\frac{1}{4\pi^2}$$

$$X(f)=\frac{1}{(f-\alpha)\cdot\left(f-\frac{3}{2}\alpha\right)}$$

Por frações parciais

$$X(f) = \frac{1}{(f-\alpha)\cdot(f-\frac{3}{2}\alpha)} = \frac{a}{f-\alpha} + \frac{b}{f-\frac{3}{2}\alpha}$$

$$a\cdot(f-\frac{3}{2}\alpha) + b(f-\alpha) = 1$$

$$f(a+b) + \alpha(-\frac{3}{2}a - b) = 1$$

$$f(a+b) = 0 \qquad a+b=0 \qquad a=-b$$

$$\alpha(-\frac{3}{2}a - b) = 1$$

$$-3a - 2b = \frac{2}{\alpha}$$

$$-3\cdot -b - 2b = \frac{2}{\alpha}$$

$$b = \frac{2}{\alpha} \qquad \alpha = \frac{j}{\pi}$$

$$b = \frac{2\pi}{j} \qquad a = -\frac{2\pi}{j}$$

$$= \frac{1}{(f-\alpha)\cdot\left(f-\frac{3}{2}\alpha\right)} = \frac{a}{f-\alpha} + \frac{b}{f-\frac{3}{2}\alpha}$$

$$= \frac{-2\pi}{\left(f-\frac{j}{\pi}\right)\cdot j} + \frac{2\pi}{\left(f-\frac{3}{2}\frac{j}{\pi}\right)\cdot j}$$

$$= 2\pi\left(\frac{-1}{\left(fj+\frac{1}{\pi}\right)} + \frac{1}{\left(fj+\frac{3}{2\pi}\right)}\right)$$

$$= 2\pi\left(\frac{-j}{\left(-f+\frac{j}{\pi}\right)} + \frac{j}{\left(-f+\frac{3j}{2\pi}\right)}\right)$$

$$= 2\pi j\left(\frac{-1}{-\left(f-\frac{j}{\pi}\right)} + \frac{1}{-\left(f-\frac{3j}{2\pi}\right)}\right)$$

$$= 2\pi j\left(\frac{1}{\left(f-\frac{j}{\pi}\right)} - \frac{1}{\left(f-\frac{3j}{2\pi}\right)}\right)$$

Voltando o P

$$P \cdot X(f) = \frac{1}{4\pi^2} \cdot 2\pi j \left(\frac{1}{(f-\alpha)} - \frac{1}{(f-\frac{3}{2}\alpha)} \right)$$

$$= \frac{j}{2\pi} \cdot \left(\frac{1}{(f-\alpha)} - \frac{1}{(f-\frac{3}{2}\alpha)} \right)$$

$$P = \frac{j}{2\pi} \qquad X(f) = \frac{1}{(f-\alpha)} - \frac{1}{(f-\frac{3}{2}\alpha)}$$

$$x(t) = P \cdot \left\{ \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-\alpha)} \cdot e^{j2\pi f t} \, df}_{1} - \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-\frac{3}{2}\alpha)} \cdot e^{j2\pi f t} \, df}_{2} \right\}$$

$$1 \quad \int_{-\infty}^{\infty} \frac{e^{j2\pi f t}}{(f-\alpha)} \, df = \int_{-\infty}^{\infty} \frac{e^{j2\pi f t}}{(f-\frac{j}{4})} \, df$$

Aplicando o Teorema dos resíduos

1- Identificação do polo

$$f = \frac{j}{\pi}$$

2- Fechamento do contorno

$t > 0$: fechamos o contorno no semiplano superior

$t < 0$: Como o polo $\left(f = \frac{j}{2\pi}\right)$ se encontra apenas no semiplano superior, a integral vale 0 e fechamos o semiplano inferior.

3- Cálculo do Resíduo

$$\text{Res}\,(f = j/\pi) = \lim_{f \to \frac{j}{\pi}} (f - j\pi) \cdot \frac{e^{j2\pi f t}}{(f - j\pi)}$$

$$= e^{j2\pi \frac{j}{\pi} t} = e^{-2t} \cdot u(t)$$

$u(t)$ adicionado, pois só é válido para $t \geq 0$.

4- Resultado

$$\int_{-\infty}^{\infty} \frac{e^{j2\pi f t}}{f - \frac{j}{\pi}} = 2\pi j \cdot Res(f = j/\pi)$$

$$= 2\pi j \cdot (e^{-2t} \cdot u(t))$$

(2) $$\int_{-\infty}^{\infty} \frac{e^{j2\pi f t}}{f - \frac{3j}{2\pi}} = 2\pi j \cdot Res(f = 3j/2\pi)$$

$$= 2\pi j \cdot e^{3t} \cdot u(t)$$

Cálculo do Resíduo

$$Res(f = 3j/2\pi) = \lim_{f \to 3j/2\pi} \left(f - \frac{3j}{2\pi}\right) \cdot \frac{e^{j2\pi f t}}{\left(f - \frac{3j}{2\pi}\right)}$$

$$= e^{j 2\pi \frac{3j}{2\pi} t} = e^{-3t} \cdot u(t)$$

$u(t)$ adicionado, pois só é valido para $t \geq 0$.

Voltando ao cálculo de $x(t)$

$$P = \frac{j}{2\pi}$$

$$x(t) = P \cdot \left\{ \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-2)} \cdot e^{j2\pi f t} df}_{1} - \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-\frac{3}{2}2)} \cdot e^{j2\pi f t} df}_{2} \right\}$$

$$x(t) = \frac{j}{2\pi} \cdot \left(2\pi j \cdot e^{-2t} \cdot u(t) - 2\pi j \cdot e^{-3t} \cdot u(t) \right)$$

$$= \frac{2\pi \cdot j^2}{2\pi} \left(e^{-2t} - e^{-3t} \right) \cdot u(t)$$

$$= \left(e^{-3t} - e^{-2t} \right) \cdot u(t)$$